

303- "Range Sum Query - Immutable"

* Given an integer array `nums`, handle multiple queries of the following type

① calculate the sum of elements of `nums` b/w index `left` and `right` ($left \leq right$)

• Implement the `NumArray` class:-

① `NumArray(int[] nums)` constructor of class initializes the object with the integer array `nums`.

② `int sumRange(int left, int right)` Return the sum of elements of `nums` b/w index `left` to `right`.

Input:- `["NumArray", "sumRange", "sumRange", "sumRange"]` → Queries

`[-2, 0, 3, -5, 2, -1]` → original Array

first query initializes the `NumArray` object

left-Right query, `[0, 2]`, `[2, 5]`, `[0, 5]`

output:- `[-1, 3]` → Expected.

Explanation/-logic:- we have to find the sum of required index kind of subarray.

Brute force:- we have to find all subarray sum then return their sum of required based. the time comp will be $\approx O(N^2)$

Date: _____

Optimal Solution:- Using prefix method.

original Array:- $[-2, 0, 3, -5, 2, -1]$ $n = \text{Array.length}$

* If Add the prefix sum in other Array than find out required passed.

PrefixSum = $[-2, -2, 1, -4, -2, -3]$

* PrefixSum[0] = Array[0];

$i = 1 - n \rightarrow \text{PrefixSum}[i] = \text{PrefixSum}[i-1] + \text{Arr}[i]$

* let's check if it give Answer ok:

let's suppose we have to 0-2 sum

Return Prefix[2] = 1

verify $\rightarrow -2 + 0 + 3 = 1 \rightarrow \text{ok}$

* But what if 2-5

If we Return Prefix[5] = -3

verify $\rightarrow 1 + (-4) + (-2) + (-3) =$

$3 + (-5) + 2 + (-1) = -1$

No + Satisfied

* Prefix[5] mean 0-5 Here 0-1 are

include right if subtract it than

it's fine 0-1 mean Prefix[1] = -2

Prefix[5] - Prefix[1] = $-3 - (-2) = -1$

Now it's fine

* Mean:- Prefix[right] - Prefix[left-1];

* problem:- If the range start from 0

left-1 error index out of range

* If we Add "0" on first index mean 0

index than it might not be problem 2.

MIGHTY PAPER PRODUCT

let's see

$$\text{PrefixSum} = [0, -2, -2, 1, -4, -2, -3]$$

* then we have to ~~subtract~~ ^{add} 1 from right index b/c we add one element on PrefixSum that's why

* If we add one extra element we don't need to ^{left} "from" Main logic = $\text{PrefixSum}[\text{right} + 1] - \text{Prefix}[\text{left}]$

let's Dry Run it:-

$$[0-5] \rightarrow -2 + 0 + 3 + (-5) + (2) + (-1) = -3$$

$$\begin{aligned} * \text{PrefixSum}[5+1] - \text{PrefixSum}[0] \\ -3 - 0 = \boxed{-3} \text{ Ans} \end{aligned}$$

$$[2-5] \rightarrow -3 + (-5) + 2 + (-1) = -1$$

$$\begin{aligned} * \text{PrefixSum}[5+1] - \text{PrefixSum}[2] \\ -3 - (-2) = \boxed{-1} \text{ Ans} \end{aligned}$$